Dr. rer. nat.
Julian Q. Kosciessa

Thomas van Aquinostraat 4, Nijmegen

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ORCID: 0000-0002-4553-2794



I am an experienced researcher working at the intersection of cognitive, computational and systems neurosciences. My work aims to improve the characterization of neural dynamics, and clarify the functional role of neural rhythms and noise in flexible cognition. My experimental research combines neuroscientific intervention and measurement techniques, and extends available methods via scientific software development.



Research Experience

Postdoctoral Research Fellow — **2022 – PRESENT**
Donders Institute for Brain, Cognition and Behaviour
Nijmegen, Netherlands

Pre-/postdoctoral Research Fellow — **2016 – 2022**
IMPRS Comp2Psych
Max Planck UCL Centre for Computational Psychiatry and Aging Research
Berlin, Germany

2010 – 2016

Research Assistant/Intern
Berlin, Germany
London, UK
Singapore, Singapore

Education

Humboldt Universität zu Berlin — **2016 – 2020**
Psychology
Dr. rer. nat. (summa cum laude)

Humboldt Universität zu Berlin — **2014 – 2016**
Mind & Brain – Track Brain
M.Sc. Master of Science

Freie Universität Berlin — **2011 – 2014**
Psychology
B.Sc. Bachelor of Science

Comp. Skills

MATLAB
R
UNIX
Python
Git

Languages

German	native
English	near-native
Chinese	HSK5
Dutch	B1
French	A2

Key Publications

	Citations
Kosciessa, J. Q., Lindenberger, U., & Garrett, D. D. (2021) Thalamocortical excitability adjustments guide human perception under uncertainty *Nature Communications*	**102**
Kosciessa, J. Q., Kloosterman, N. A., & Garrett, D. D. (2020) Standard multiscale entropy reflects neural dynamics at mismatched temporal scales: What's signal irregularity got to do with it? *PLoS Computational Biology*	**104**
Kosciessa, J. Q., Grandy, T. H., Garrett, D. D., & Werkle-Bergner, M. (2020) Single-trial characterization of neural rhythms: Potential and challenges. *NeuroImage*	**122**

RESEARCH

11/2023 - PRESENT **Research Fellow**
Donders Centre for Cognition
Radboud University, Nijmegen, The Netherlands
Cognitive Neurodynamics

09/2022 – PRESENT **Postdoctoral Researcher | Radboud Excellence Fellow**
Donders Institute for Brain, Cognition and Behaviour
Radboud University, Nijmegen, The Netherlands
Cognitive Neuromodulation | PI: Dr. Lennart Verhagen

07/2020 – 07/2022 **Postdoctoral Researcher**
Max Planck Institute for Human Development, Berlin, Germany

10/2016 – 03/2020 **Predoctoral Research Fellow**
IMPRS COMP2PSYCH: International Max Planck Research School on Computational Methods in Psychiatry and Ageing Research
Max Planck UCL Centre for Computational Psychiatry and Aging Research
Max Planck Institute for Human Development, Berlin, Germany
Supervisors: Prof. Dr. Ulman Lindenberger, Dr. Douglas D. Garrett

10/2015 – 03/2016 **Research Intern**
UCL Institute of Cognitive Neuroscience
PIs: Prof. Emrah Düzel & Prof. Ray Dolan
PI: Dr. Dorothea Hämmerer

03/2015 – 07/2015 **Research Intern**
Max Planck Institute for Human Development, Berlin, Germany
Center for Adaptive Rationality (ARC)
PI: Dr. Wouter van den Bos

09/2012 – 09/2013 **Research Assistant**
07/2014 – 09/2015 Max Planck Institute for Human Development, Berlin, Germany
04/2016 – 09/2016 Cognitive and neuronal dynamics of memory across the lifespan
PIs: Dr. Markus Werkle-Bergner & Dr. Yee Lee Shing

01/2014 – 05/2014 **Research Intern**
Cognitive Neuroscience Laboratory, Duke-NUS, Singapore
PI: Prof. Michael Chee
Supervisor: Dr. Irma Kurniawan

EDUCATION

10/2016 – 10/2020 **Humboldt Universität zu Berlin**
Psychology. Dr. rer. nat. (summa cum laude)

10/2014 – 09/2016 **Humboldt Universität zu Berlin**
Mind & Brain – Track Brain. M.Sc. Master of Science (GPA: 1.0)

09/2015 – 04/2016 **University College London**
Two Erasmus exchange terms. Institute of Neurology

07/2013 – 05/2014 **National University of Singapore (NUS)**
Two exchange semesters. Faculty of Arts and Social Sciences

10/2011 – 09/2014 **Freie Universität Berlin**
Psychology. B.Sc. Bachelor of Science (GPA: 1.1)

Teaching

2023–2025 | Lab Rotations, MSc Cognitive Neuroscience, Radboud University, The Netherlands
2023–2025 | Practical Training and Thesis. MSc Cognitive Neuroscience, Radboud University, The Netherlands
2025 | Lecture: Flexible Perceptual and Cognitive Control. Trends in Cognitive Neuroscience, MSc Cognitive Neuroscience, Radboud University, The Netherlands
2025 | Lecture: Managing Uncertainty in Perceptual and Cognitive Control. Ghent University, Belgium
2024 | Lecture: Evolving Models of Flexible Neuro-Cognitive Control. Trends in Cognitive Neuroscience, MSc Cognitive Neuroscience, Radboud University, The Netherlands
2024-2025 | Workshop: Multimodal TUS-fMRI. Donders Toolkit Advanced (f)MRI. Radboud University. The Netherlands
2023 | Workshop: Managing Your Data with DataLad. CuttingEEG, Frankfurt, Germany
2023–2025 | Workshop: Simulations for Transcranial Ultrasound Stimulation. NeuroTechEU Course / Donders NIBS Toolkit. Radboud University. The Netherlands [link]
2023 | Seminar Teacher: Higher-Order Cognition and Emotion. Radboud University Medical Center, The Netherlands
2023 | Psychology Research Project 3. Teacher. Radboud University
2022 | Workshops: Reusable data management with DataLad.
Cognitive Psychology. University of Münster. Germany
Donders Centre for Cognitive Neuoimaging. Radboud University. The Netherlands
2020 | Workshop: Multi-scale entropy as a tool to characterize neural signal irregularity. EEG Meeting. Max Planck Institute for Human Development. Berlin, Germany
2018 | Seminar: Methods for the analysis of rhythmic and arrhythmic brain activity.International Max Planck Research School on the Life Course. Berlin, Germany

Talks (selected)

2025 | Invited Symposium Talk: Neuromodulatory Influences On Aging Cognition: Novel Measurement And Intervention Approaches In Health And Disease. ICON2025. Porto, Portugal
2025 | Talk: Biasing perceptual and cognitive control with thalamic ultrasound stimulation. Focused Ultrasound Neuromodulation Conference (FUN25), Hong Kong
2024 | Talk: Neuroscientific data management using DataLad. distribits conference. Düsseldorf. Germany [link]
2023 | Invited Keynote Talk: Signs, signals, and noise in human brain dynamics. CuttingEEG conference. Frankfurt. Germany [link]
2022 | Invited Research Talks: Dynamic neural regimes for flexible decisions under uncertainty.
Translational Decision-Making Seminar. University of Minnesota/Université de Montréal. U.S.A./Canada
Biopsychology. University of Munster. Germany
Donders Institute for Brain, Cognition and Behaviour. The Netherlands
2022 | Invited Symposium Talk: Influences of arousal and cortical excitability on adaptive perceptual decision making. ICON 2020. Helsinki, Finland
2021 | Research Talk: The role of neural dynamics in flexible perception under uncertainty. Computational Neuroscience Symposium. Osnabrück, Germany
2021 | Invited Research Talks: Thalamocortical excitability adjustments guide human perception under uncertainty. Shine Lab, University of Sydney, Australia / Halassa Lab, Massachusetts Institute of Technology (MIT), U.S.A.
2020 | Invited Colloqium Talk: Measurement and relevance of rhythmic and aperiodic human brain dynamics. Technical University. Berlin, Germany

Fellowships

2022 – 2024 Radboud Excellence Initiative Fellowship
Donders Centre for Cognition, Radboud University, The Netherlands
2022 ReproNim/INCF Fellowship (https://repronim.org/fellowship/)
2016 – 2020 IMPRS COMP2PSYCH Fellowship
International Max Planck Research School on Computational Methods in Psychiatry and Ageing Research, Germany

Awards

2025 | Poster Award (Gold) | Focused Ultrasound Neuromodulation Conference (FUN25)
2022 | Radboud Excellence Initiative Fellowship (200.000 EUR)
2022 | Otto Hahn Medal of the Max Planck Society (7.500 EUR)
2022 | Conference Travel Grant | DAAD | International Conference of Cognitive Neuroscience
2021 | DGPA Brain Products Young Scientist Award 2021
2021 | Conference Travel Grant | DAAD | OHBM Meeting 2021
2021 | Merit Abstract Award | OHBM Meeting 2021
2018 | IBRO Poster Award | Interpreting BOLD 2018
2018 | Conference Travel Grant | DAAD | Interpreting BOLD 2018 (Oxford, UK)
2015 | Erasmus Stipend (2015/16) | DAAD | University College London, UK
2014 | PROMOS Stipend | DAAD | National University Singapore, Singapore

Mentorship

2025 | Sebastian Reichstein | Research Assistant (MSc)
2025 | Ruggero Condorelli | Research Assistant (MSc)

2025/26 | Narges Naghibi | MSc Cognitive Neuropsychology, Vrije Universiteit Amsterdam
2025/26 | Lena Grondelaers | MSc Cognitive Neuroscience, Radboud University
2025/26 | Javier Gómez Pérez | MSc Cognitive Neuroscience, Radboud University
2024/25 | Joana Pedroso de Faria | MSc Cognitive Neuroscience, Radboud University
2024/25 | Sebastian Reichstein | MSc Cognitive Neuroscience, Radboud University
2023/24 | Martin Wimmers | MSc Cognitive Neuroscience, Radboud University
2025+ PhD student at TU Eindhoven
2023 | Jesse Lam | MSc Cognitive Neuroscience, Radboud University
2023 | Lieke Hendrix | BSc Biology, Radboud University
2023 | Sara Mulders | BSc Psychology, Radboud University
2023 | Siem van der Sluijs | BSc Psychology, Radboud University
2021/22 | Claire Pleche | M.Sc. Cognitive Neuroscience, Ecole Normale Supérieure de Paris, France
2022+ PhD student at Donders Institute
2021-2023 | Mentor at Neuromatch Academy

Ultrasound Trials *(CCMO)*

NL86115.091.24 | Biasing decision flexibility with thalamic ultrasound stimulation [link]
NL83639.091.23 | Imaging deep target engagement of transcranial ultrasonic stimulation [link]

Professional Activites & Leadership

Ad-hoc peer reviewer: *PNAS, TICS, PLoS Biology, NeuroImage (10x), Journal of Neuroscience, Journal of Neurophysiology, International Journal of Psychophysiology, Brain Topography, European Journal of Neuroscience, Mindfulness, PLoS One* (2x)

2018+ | Organization for Human Brain Mapping (OHBM) | Member
2020+ | Deutsche Gesellschaft für Psychology (DGPs) | Associate Member
2021+ | International Neuroinformatics Coordinating Facility (INCF) | Member
2022+ | International Transcranial Ultrasonic Stimulation Safety and Standards (ITRUSST) | Member
2022+ | Co-founder & co-organizer: "DCC Nexus" meeting series
2022-2024 | PostDoc representative at the Donders Centre for Cognition
2022-2024 | Donders Institute PostDoc Council | Member
2023 | Co-founder & -organizer Donders PostDoc Ponders
2024 | Hiring Committee PostDoc Coordinator | Member
2025+ | ITRUSST working group on Treatment Planning | Member
2025 | AAPG [French National Research Agency] (CE37: Integrative and Cognitive Neuroscience) | Reviewer
2025 | Donders Strategy Evaluation (2020-2024) | Postdoc representative
2025 | Cognitive Computational Neuroscience (CCN) | Communications Committee (Member)
2026 | Cognitive Computational Neuroscience (CCN) | Communications Committee (Chair)

Publications (*corresponding author)

3 Skowron, A., **Kosciessa, J. Q.**, Lorenz, R., Hertwig, R., van den Bos, W., & Garrett, D. D. (in press). Neural variability compresses with increasing belief precision during Bayesian inference. Cerebral Cortex.

15 Mooraj, Z., Salami, A., Campbell, K. L., Dahl, M. J., **Kosciessa, J. Q.**, Nassar, M. R., Werkle-Bergner, M., Craik, F. I. M., Lindenberger, U., Mayr, U., Rajah, M. N., Raz, N., Nyberg, L., & Garrett, D. D. (2025). Toward a functional future for the cognitive neuroscience of human aging. Neuron, 113(1), 154-183.

3 **Kosciessa, J. Q.*,** Mayr, U., Lindenberger, U., & Garrett, D. D. (2024). Broadscale dampening of uncertainty adjustment in the aging brain. Nature Communications, 15(1), 10717.

102 **Kosciessa, J. Q.***, Lindenberger, U., & Garrett, D. D. (2021). Thalamocortical excitability adjustments guide human perception under uncertainty. *Nature Communications, 12(1), 2430*.

32 Kloosterman, N. A., **Kosciessa, J. Q.**, Lindenberger, U., Fahrenfort, J. J., & Garrett, D.D. (2020). Boosts in brain signal variability track liberal shifts in decision bias. *Elife, 9*.

104 **Kosciessa, J. Q.***, Kloosterman, N. A., & Garrett, D. D. (2020). Standard multiscale entropy reflects neural dynamics at mismatched temporal scales: What's signal irregularity got to do with it? *PLoS Computational Biology, 16(5)*.

122 **Kosciessa, J. Q.***, Grandy, T. H., Garrett, D. D., & Werkle-Bergner, M. (2020). Single-trial characterization of neural rhythms: Potential and challenges. *NeuroImage, 206, 116331*.

172 Hämmerer, D., Callaghan, M. F., Hopkins, A., **Kosciessa, J.**, Betts, M., Cardenas-Blanco, A., Kanowski, M., Weiskopf, N., Dayan, P., Dolan, R. J., & Düzel, E. (2018). Locus coeruleus integrity in old age is selectively related to memories linked with salient negative events. Proceedings *of the National Academy of Sciences of the United States of America, 115, 2228-2233*.

Software / Tools (selected)

PRESTUS | PREprocessing & Simulations for Transcranial Ultrasound Stimulation | 10.5281/zenodo.15095860
eBOSC | extended Better OSCillation Detection | 10.5281/zenodo.4668501
eBOSC (python) | extended Better OSCillation Detection (python version) | 10.5281/zenodo.5898812
mMSE | modified multiscale entropy | 10.5281/zenodo.4672137
MAAT | Multi-attribute attention task | 10.5281/zenodo.14216065

PUBLICATION LIST

Journal Publications *(*corresponding author)*

- Skowron, A., **Kosciessa, J. Q.**, Lorenz, R. C., Hertwig, R., van den Bos, W., & Garrett, D. D. (2025). Neural variability in the default mode network compresses with increasing belief precision during Bayesian inference. Cerebral Cortex, 35(9).

  BOLD variability tracks changes in internal uncertainty levels during a latent inference task.

  3 #
  article

- Mooraj, Z., Salami, A., Campbell, K. L., Dahl, M. J., **Kosciessa, J. Q.**, Nassar, M. R., Werkle-Bergner, M., Craik, F. I. M., Lindenberger, U., Mayr, U., Rajah, M. N., Raz, N., Nyberg, L., & Garrett, D. D. (2025). Toward a functional future for the cognitive neuroscience of human aging. Neuron, 113(1), 154-183.

  A perspective on dominant methodologies in the cognitive neuroscience of human aging, identifying an overreliance of structural and resting-state measures. The perspective advocates for a more task-centric, computationally-informed, approach in the field.

  15 #
  article

- **Kosciessa, J. Q.*,** Mayr, U., Lindenberger, U., & Garrett, D. D. (2024). Broadscale dampening of uncertainty adjustment in the aging brain. Nature Communications, 15(1), 10717.

  Task uncertainty is tracked by an integrated neural response (EEG & fMRI-based) across 100 younger and older adults, including prefrontal-thalamic modulation. This response dampens in older age, while response maintenance is associated with higher cognitive flexibility.

  3 #
  article
  data
  code

- **Kosciessa, J. Q.***, Lindenberger, U., & Garrett, D. D. (2021). Thalamocortical excitability adjustments guide human perception under uncertainty. *Nature Communications, 12(1), 2430*.

  Higher-order thalamic activation increases when contextual uncertainty ambiguates which environmental features are critical for an upcoming choice, and is associated with switches from a rhythmic to an aperiodic processing mode.

  102 #
  article
  data
  code
  task
  press release

- Kloosterman, N. A., **Kosciessa, J. Q.**, Lindenberger, U., Fahrenfort, J. J., & Garrett, D.D. (2020). Boosts in brain signal variability track liberal shifts in decision bias. *Elife, 9*.

  The magnitude of adaptive shifts from conservative to liberal decision biases under speed-accuracy emphasis is tracked by increasing signal variability in frontal cortex.

  32 #
  article
  data
  code

- **Kosciessa, J. Q.***, Kloosterman, N. A., & Garrett, D. D. (2020). Standard multiscale entropy reflects neural dynamics at mismatched temporal scales: What's signal irregularity got to do with it? *PLoS Computational Biology, 16(5)*.

  Highlights and exemplifies biases in prior research using an information theoretic metric of signal irregularity and proposes avenues to adjudicate such issues in future applications.

104 #
article
data
code
toolbox

- **Kosciessa, J. Q.***, Grandy, T. H., Garrett, D. D., & Werkle-Bergner, M. (2020). Single-trial characterization of neural rhythms: Potential and challenges. *NeuroImage, 206, 116331*.

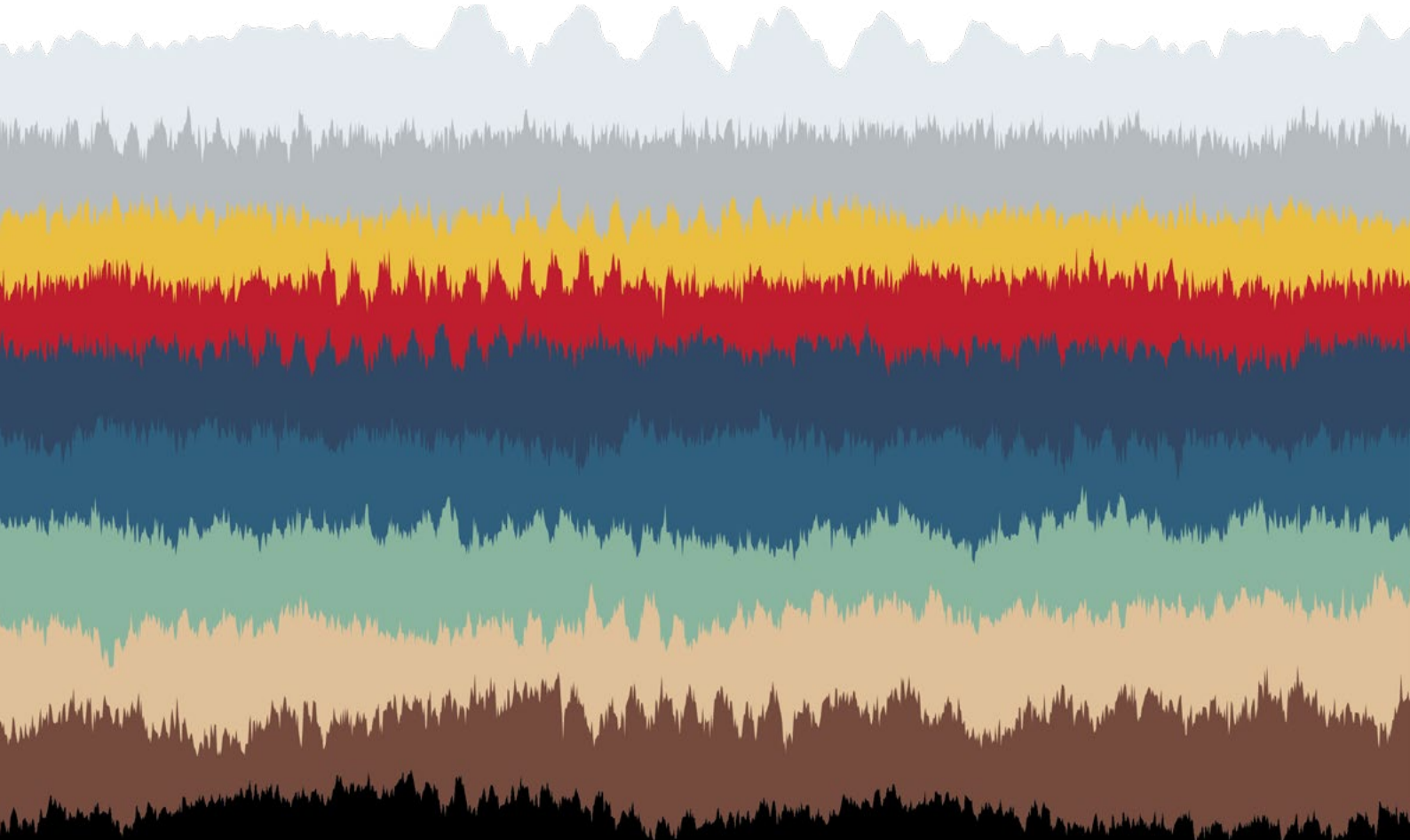
  Introduces a novel algorithm that separates neural rhythms from background activity in magnitude, space and time, and enables specific rhythm characterization when boundary conditions are met.

122 #
article
code
toolbox

- Hämmerer, D., Callaghan, M. F., Hopkins, A., **Kosciessa, J.**, Betts, M., Cardenas-Blanco, A., Kanowski, M., Weiskopf, N., Dayan, P., Dolan, R. J., & Düzel, E. (2018). Locus coeruleus integrity in old age is selectively related to memories linked with salient negative events. *Proceedings of the National Academy of Sciences of the United States of America, 115, 2228-2233*.

  Quantitative imaging indicates structural reductions in brainstem locus coeruleus integrity with increasing adult age, and links related noradrenergic drive to the encoding of salient events.

172 #
article



Monographs/Theses

- **Kosciessa, J. Q.** (2020, Dr. rer. nat.). Measurement and relevance of rhythmic and aperiodic human brain dynamics. *Humboldt-Universität zu Berlin*.

  This dissertation highlights improvements in the ability to selectively characterize rhythmic and aperiodic fluctuations, and discusses potential generating mechanisms as well as modulatory influences to contextualize their interpretation at the latent level of human brain function.

  article

- **Kosciessa, J. Q.** (2016, M. Sc.). Effects of short-term memory load and task training on the amplitude and abundance of rhythmic neural activity. *Humboldt-Universität zu Berlin*

  Neural rhythms are dominantly characterized by their power, but this measure conflates amplitude and duration. This work separates these two parameters of human alpha rhythms and investigates how they are modulated during working memory.

- **Kosciessa, J.** (2014, B. Sc.). The assessment of microsaccades from the rEOG. *Freie Universität Berlin*

  Microsaccades are small, high-velocity eye movements. This work explores the potential to use visual EEG channels to detect microsaccades without an eye tracker, and describes adult age differences in microsaccade characteristics.

Preprints / in preparation

- **Kosciessa, J. Q.*** (2022). EEGmanylabs contribution.

  Preprocessing and analysis workflows for high-dimensional EEG data are highly variable. This project aims to establish the impact of analytical variability on hypothesis testing. This is a contribution to this community effort that was derived with one of my common workflows.

  preprint | code + data

- Garrett, D. D., Kloosterman, N. A., Epp, S., Chopurian, V., **Kosciessa, J. Q.**, Waschke, L., Skowron, A., Shine, J. M., Perry, A., Salami, A., Rieckmann, A., Papenberg, G., Wåhlin, A., Karalija, N., Andersson, M., Riklund, K., Lövdén, M., Bäckman, L., Nyberg, L., & Lindenberger, U. (2022). Dynamic regulation of neural variability during working memory reflects dopamine, functional integration, and decision-making. bioRxiv, 2022.2005.2005.490687. https://doi.org/10.1101/2022.05.05.490687

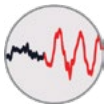
  Linking BOLD variability increases with increasing memory load to dopamine capacity, network-level functional integration, and flexible decision processes, this study argues that the ability to dynamically regulate subcortical striato-thalamic dynamics according to momentary task demands may be a hallmark of adaptive brain function.

  preprint

- **Kosciessa, J. Q.,** Martin Wimmers, Sophie Krijgsman, Lennart Verhagen. (2025). Dissociable Effects of Uncertainty in Perceptual and Cognitive Control. CCN 2025.

Software / Tools / Data

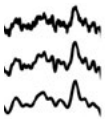
(abridged; see https://jkosciessa.github.io/tools for complete and updated list)



extended Better OSCillation Detection (**eBOSC**)

Toolbox to detect the occurrence of rhythms in (neural) time series. *MATLAB/Python*

DOI 10.5281/zenodo.4668502
DOI 10.5281/zenodo.5898813



modified multiscale entropy (**mMSE**)

Information-theoretic algortihm for the characterization of (neural) time series irregularity.

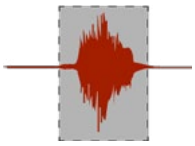
DOI 10.5281/zenodo.4672138



Multi-attribute attention task (**MAAT**)

Adaptation of the random dot motion paradigm, a key task in systems neuroscience.

DOI 10.5281/zenodo.14216065



Computer-Assisted Response Labeler (**CARL**)

Graphical User Interface that accelerates verbal response labelling in psychological experiments.

DOI 10.5281/zenodo.7505622



PREprocessing & Simulations for Transcranial Ultrasound Stimulation (**PRESTUS**)

Toolbox to create personalized models of ultrasonic brain stimulation based on structural MRI images.

DOI 10.5281/zenodo.15095860

Examples of integrated code & data DataLad datasets:
https://gin.g-node.org/juliankosciessa/eegmp_analysis
https://git.mpib-berlin.mpg.de/LNDG/stateswitchage/stsw
https://gin.g-node.org/StateSwitch/stsw



see also:
https://git.mpib-berlin.mpg.de/LNDG/stateswitch
https://osf.io/ug4b8/